

Ankur Aditya

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RESEARCH INTERESTS

Multimedia Systems, Computer Vision, RF/Wireless Sensing

EDUCATION

University of Massachusetts (UMass), Amherst

Doctor of Philosophy (MS/PhD), Computer Science

Relevant Courseworks: Distributed and Operating Systems, Intelligent Visual Computing, Advanced Algorithms.

Sept '22 - Present

CGPA: 3.7/4

Indian Institute of Technology Kanpur (IIT-K), Kanpur

Doctor of Philosophy (MS/PhD), Electrical Engineering

Transferred to UMass Amherst for continuation of Ph.D.

Relevant Courseworks: MIMO Wireless Communications, Detection and Estimation Theory.

Jan '21 - May '22

CGPA: 8.67/10

International Institute of Information Technology (IIIT), Bhubaneswar

Bachelor of Technology (B.Tech), Electronics and Telecommunication Engineering

Relevant Courseworks: Antenna and Wave Propagation, Mobile Communications.

Aug '16 - June '20

CGPA: 9.06/10

PUBLICATIONS

- X. Chen, **Ankur Aditya**, Z. Lei, D. Ganesan. 2023. Curtain-Net: Enabling precise beamforming with a deformable antenna array on a fabric substrate. In The 21st ACM Conference on Embedded Networked Sensor Systems (SenSys '23), November 12–17, 2023, Istanbul, Turkiye. ACM, New York, NY, USA, 13 pages.
- **Ankur Aditya** et al., D2RF: Beamforming with Flexible Antenna on a Textile Substrate (**Under Review**).

EXPERIENCE

Research Assistant, CICS UMass

Advisor: [Prof. Prashant Shenoy](#)

Amherst, MA, USA

May '25 - Present

• Project: Loss Resilient Volumetric Video Streaming System

- Developing a novel application layer loss recovery for recovering corrupted frames in volumetric video streaming, improving reliability under network packet loss conditions.
- Designed a two-stage learning pipeline leveraging Vision Transformer (ViT) models at the client-end to enhance temporal consistency and robustly reconstruct corrupted frames from compressed video streams.
- Achieved real-time frame recovery with negligible impact on rendering latency, demonstrating a significant improvement in quality (PSNR/SSIM) over baseline error concealment methods.

Research Assistant, CICS UMass

Advisor: [Prof. Deepak Ganesan](#)

Amherst, MA, USA

Jan '24 - Aug '24

• Project: D2RF: Beamforming with Flexible Antenna on a Textile Substrate (**Under Review**)

- Developed a novel predictive model using a Vision Transformer (ViT) to estimate the electromagnetic (EM) characteristics of flexible antennas.
- Demonstrated the successful application of computer vision methods to a real-world physics problem, achieving comparable accuracy to EM simulators (e.g., HFSS) but orders of magnitude faster.
- Built a comprehensive simulation library (D2EM) mapping physical deformations to EM properties, enabling a 2x improvement in physical beamforming and over 20x in large-scale simulations.

Research Assistant, CICS UMass

Advisor: [Prof. Deepak Ganesan](#)

Amherst, MA, USA

Jan '23 - May '23

• Project: Curtain-Net: Enabling precise beamforming with a deformable antenna array on a fabric substrate

- Explored solutions for signal degradation caused by physical deformations (curvature, bending) in flexible antenna arrays mounted on fabric.
- Developed CurtainNet, an end to end system delivering over 12dB of beamforming gain and extending wireless coverage radius by up to 20m by correcting for these deformations.

Teaching Assistant, CICS UMass

Courses:

- Reasoning Under Uncertainty (Fall '22, Fall '23, Spring '25 and Fall '25)
- Introduction to Computation (Fall '24)
- Scalable Web Systems (Summer '23)

Amherst, MA, USA

Research Intern, EE IIT-K

Advisor: [Prof. K. V. Srivastava](#)

Kanpur, UP, India

Dec '19 - June '20

• Project: Design of Microwave Absorbers with Transmission Band using Frequency Selective Surfaces

- Designed and simulated a novel, ultra-thin (5.57mm) metamaterial absorber with an integrated transmission band, achieving a peak insertion loss of only 0.167dB.

SKILLS

- **Programming Languages:** ◇ Python ◇ MATLAB
- **Frameworks & Libraries:** ◇ PyTorch ◇ NumPy ◇ Open3D ◇ scikit-learn ◇ TorchCodec
- **Antenna Simulation Tools:** ◇ Ansys HFSS ◇ CST Microwave Studio
- **Other Tools:** ◇ Git ◇ GNU Radio ◇ Blender ◇ PyRealSense ◇ LaTeX